

Prince Gulia

Backend Developer

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PROFILE

Backend Developer and final-year BCA student specialising in scalable REST APIs, real-time architectures, and AI-integrated systems. Hands-on experience building asynchronous background job queues (BullMQ/Redis), JWT authentication systems, file-processing pipelines, and secure document-grounded RAG pipelines using pgvector and Gemini API. Focused on backend efficiency, scalability, and building production-ready backend systems.

EDUCATION

Bachelor of Computer Applications (BCA) - Guru Gobind Singh Indraprastha University (IINTM, Janakpuri)

2024 – 2027 (Expected) | New Delhi, India | **CGPA: 9.52 / 10.0**

- Coursework: Data Structures & Algorithms, DBMS, Web Development, Operating Systems

TECHNICAL SKILLS

Languages - JavaScript, C++, SQL, Python (academic)

Backend - Node.js, Express.js, RESTful API Design, WebSockets, Socket.io, JWT Authentication, Bcrypt

Databases - PostgreSQL, pgvector, Supabase, MySQL, SQLite

Async / Infra - Redis, BullMQ, Job Queues

AI / ML - RAG Pipelines, Vector Embeddings, Gemini API, Semantic Search

File Handling - Multer, Cloudinary, Sharp (Image Processing)

Deployment - Render, Vercel, Supabase (hosted PostgreSQL)

Tools - Git, GitHub, Postman, Linux

PROJECTS

DocuMind - AI Document Assistant

Node.js, Express, PostgreSQL, pgvector, BullMQ, Gemini API, RAG [Code](#) | [Live Demo](#)

- Architected a RAG pipeline using Gemini API for vector embeddings and pgvector for cosine similarity search across document chunks.
- Engineered an asynchronous PDF processing queue using BullMQ and ioredis so uploads are handled in the background without blocking API requests.
- Improved retrieval quality by designing a custom 500-word chunking strategy with boundary overlap, reducing context fragmentation across chunk splits.
- Designed strict AI query guardrails to guarantee document-grounded responses and eliminate out-of-scope hallucinations from the LLM.

File Processing API

Node.js, Express, BullMQ, Redis, Multer, Cloudinary, Sharp [Code](#) | [Live Demo](#)

- Built an async file ingestion API handling multi-format uploads (images, PDFs) via Multer with queue-based background processing using BullMQ and Upstash Redis.
- Integrated Sharp for server-side image transformation (resize, compress, format conversion) and Cloudinary for cloud media storage and CDN delivery.
- Implemented job status polling endpoints allowing clients to track processing state (pending, processing, completed) without blocking the main request thread.
- Configured BullMQ with exponential backoff retry logic and maxRetriesPerRequest: null for reliable Redis connection handling under Render's hosted environment constraints.

RealChat - Real-Time Chat Application

Node.js, Express, Socket.io, PostgreSQL, JWT [Code](#) | [Live Demo](#)

- Built a real-time chat backend with Socket.io handling persistent WebSocket connections authenticated via JWT, supporting room-based messaging with accurate user-room mapping.
- Optimised payload structure and connection event handling for fast message delivery in a low-resource deployment environment.
- Engineered a PostgreSQL schema for message history with B-tree indexing on frequently queried columns to speed up chat history retrieval.
- Implemented secure room-based broadcasting isolating chat events per room and preventing cross-room data leakage.

INTERESTS

Backend system design, AI/LLM integration, distributed job queues, semantic search architectures.